

Theta-Resolved Optimal Truncation of q -Multinomial Transseries

A phase-dependent $\sqrt{x}$ shift, arithmetic error actions,
and an exact hierarchy of pole-resolved approximations

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Abstract

The positive-remainder approach to q -multinomial asymptotics leaves a specific question: how does a continuous least-remainder saddle turn into an arithmetic lattice saddle? We give a uniform, all-order answer when the lattice spacing is bounded in units of the saddle width. More importantly, we distinguish a conventional least-term rule from the *actual* minimization over all Bernoulli truncation orders. The latter is governed by a tilted Gaussian lattice sum. If $z = a_*x$, $\beta = 2\pi$, $\lambda = h\sqrt{z/\beta}$ and $\theta = \{\beta/h\}$, then

$$2K_{\min} = \beta z + \eta_*(\lambda, \theta)\sqrt{\beta z} + O(1),$$

where η_* is the unique tilt making the lattice mean zero. The minimum remainder has leading value $\frac{m_*\sqrt{2\pi}}{\beta^{3/2}\sqrt{z}}e^{-\beta z} \min_{\eta} \mathcal{J}_{\lambda}(\theta, \eta)$. We also calculate the bounded correction to the continuously optimal order and the first correction to the minimum itself.

For fixed nonresonant $0 < h < \beta$, the optimal error instead has an exact arithmetic action $I_h > \beta$, determined by the two lattice points straddling β . Its prefactor retains an oscillation from the integer truncation order. A mesoscopic theorem matches this two-point law to the large-mesh theta law. Finally, resolving the first L partial-fraction poles exactly replaces β by $2\pi(L+1)$ throughout the critical-mesh theory. Explicit positive tail bounds turn these representations into finite forward and inverse error certificates.

The article supplies conventional proofs, symbolic checks, and numerical diagnostics. The claims are scoped extensions of the inspected ProveIt transseries program and of an earlier research draft, not a claim to have settled general resurgence or established literature-wide priority.

Status. Unrefereed mathematical development. No Lean build, interval-arithmetic validation, or independent peer review is claimed.

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1 Research target, contribution, and source boundary

1.1 The problem being advanced

The ProveIt series-and-transseries corpus develops polynomial–logarithmic calculus, exact-core inversion, and asymptotic expansions of rapidly growing combinatorial families. Its companion on combinatorial transseries includes Gaussian coefficients and the singular $q \rightarrow 1$ transition [1, 2, 3]. A preceding research draft, *Uniform q -Multinomial Transseries and Certified Inversion* [4], obtains a positive exact remainder for every positive multinomial ray. Its final section explicitly proposes two further targets: a lattice-to-integral transition formula, including the fractional saddle position, and a first resolved beyond-all-orders correction. Those are the immediate targets here.

The distinction between the sources matters. The preceding draft was read from the user’s saved research files; it is not being represented as a verified publication or as a file proved to be present in the current repository. The repository’s directly retrieved transseries READMEs were pinned at commit `ae28ea2db3c01a0b777cffa6295b8fe9c4628f27`. Current directory listings were also inspected, but the large canonical source exceeded the available file endpoint. Therefore this is a targeted extension, not an exhaustive nonduplication audit of the entire current repository. The exact mathematical starting formulas are restated and proved below, so the new theorems do not depend on trusting the earlier draft’s correctness.

The previous two sharp special cases have an instructive contrast. At $h = 0$, the ordinary near-least-term remainder has a factor $z^{-1/2}$. At $h = 2\pi/m$, a lattice point lands exactly on the saddle and that factor is absent. Simply replacing a sum by an integral misses this difference. The appropriate intermediate object is a Gaussian lattice sum rather than a Gaussian integral.

There is a second distinction, not resolved merely by finding that sum: minimizing a convenient envelope is not the same as minimizing the exact remainder. At a nonsymmetric lattice phase, an order displacement of size $\sqrt{z}$ changes the relative weights of the nearby lattice points by an order-one amount. Consequently a rule $2K = 2\pi z + O(1)$ can have the right exponential scale and still miss the best leading constant. At fixed nonresonant h , it can even miss the optimal exponential action.

Editorial note (ProveIt, 2026-09-29). The preceding draft [4] is now filed in the repository as `Uniform_q_Multinomial_Certified_Inversion/uniform_q_multinomial_transseries.tex` [12]. As delivered it had the 1,521 lines and the four labels that this article names; at its questions Q1 and Q2 (section “Further research questions and topics”) an editorial note now points here and states the scope of the answers given here. This article belongs to a unit of five packages that answer the companion’s $q \rightarrow 1$ gap for the same interpolation and are to be merged into its chapter on the singular transition [12]: the preceding draft, the central Gaussian-binomial, certified-inversion and gamma-core articles, and this article. The five write the interpolation in different normalizations ($q = e^h$ or $Q = e^{-h}$; remainders R_K , r_K , R_M or $L_h - F_{h,K}$; smallest ray parameter a_* , α or $a_{\min}$); a normalization dictionary is left to the merge. The same restriction $\tau \geq \tau_0$ appears in Theorem `q3:thm:double-scaling` of the Fabius tree’s Gaussian-coefficient volume [13], which the preceding draft’s uniform theorem removes; neither article cites it. Theorems 5.2 and 8.1 are sharp, theta-resolved instances of the canonical volume’s least-term theorem `p0:thm:optimal-truncation` [14], and Theorem 9.3 uses the residual-to-root step that the Lean theorem `Fabius.exists_eq_in_residual_interval` [15] checks in abstract form; this article cites neither.

1.2 What is new in the present development

The central results are the following, in increasing order of refinement.

- (i) Theorem 4.1 gives an all-order expansion of the exact remainder, uniform for bounded saddle-scaled mesh and bounded order tilt.
- (ii) Theorem 5.2 minimizes over *all* integer orders. It determines the $\sqrt{z}$ displacement, its bounded correction, and the first correction to the minimum error.
- (iii) Theorem 6.1 identifies the exact fixed-mesh action and the remaining integer-order oscillation. Theorem 7.2 rigorously matches it with the theta description in an overlapping regime.
- (iv) Theorem 8.1 applies the same analysis after any fixed number of poles have been resolved exactly. Theorem 9.1 supplies finite evaluation bounds; Theorem 9.3 transports them to inverse errors.

The words “all-order” always refer to a finite-order asymptotic statement for every fixed requested order. They do not assert convergence of the infinite saddle expansion.

1.3 Classical mechanisms and the novelty claim

The ingredients are not new in isolation. The partial fractions of coth, Binet’s integral, q -gamma limits, and theta transformations are classical [5, 6, 7, 8]. Discrete Laplace methods have a substantial history; a recent systematic treatment is due to Hughes, Helton, and Schlosser [9]. The resurgence of Euler–Maclaurin summation and quantum dilogarithms has been studied by Costin–Garoufalidis and Garoufalidis–Kashaev [10, 11].

The contribution advanced here is the *specific quantified package*: the positive q -multinomial remainder, its phase-dependent global minimization, the arithmetic action and order oscillation, and the reusable pole-resolved hierarchy. We do not claim priority for theta asymptotics, optimal truncation, or resurgence as general subjects. The literature check identifies relevant precedents but does not establish that no equivalent specialized theorem exists elsewhere. None of the new proofs uses a hypothesis of general Borel summability.

2 The analytic family and its exact positive remainder

Fix $r \geq 2$ and a positive ray $\mathbf{a} = (a_1, \dots, a_r)$. Throughout, set

$$A = \sum_{j=1}^r a_j, \quad d = r - 1, \quad \kappa = \frac{A^2 - \sum_j a_j^2}{2}, \quad a_* = \min_j a_j, \quad m_* = \#\{j : a_j = a_*\}, \quad z = a_* x. \quad (2.1)$$

In every uniform assertion the ray is fixed. Uniformity as its entries coalesce or tend to zero is a separate problem.

For $h > 0$, define

$$\mathcal{P}(h) = \prod_{n \geq 1} (1 - e^{-hn}), \quad (2.2)$$

$$\mathcal{A}_h(s) = \sum_{m \geq 1} \frac{e^{-ms}}{m(e^{hm} - 1)}, \quad s > 0, \quad (2.3)$$

$$F_h(x) = \kappa h x^2 - d \log \mathcal{P}(h) + \mathcal{A}_h(Ahx) - \sum_j \mathcal{A}_h(a_j hx), \quad x > 0. \quad (2.4)$$

At $h = 0$, put

$$F_0(x) = \log \Gamma(Ax + 1) - \sum_j \log \Gamma(a_j x + 1). \quad (2.5)$$

When all $a_j x$ are integers, $\exp F_h(x)$ is the Gaussian multinomial coefficient at $q = e^h$. For general ray entries, (2.4) specifies the interpolation being studied. There is no assertion about arbitrary interpolations of the same integer sequence.

To check the endpoint, let $Q = e^{-h}$. The product definition of Γ_Q gives

$$e^{F_h(x)} = e^{\kappa h x^2} \frac{\Gamma_Q(Ax + 1)}{\prod_j \Gamma_Q(a_j x + 1)}. \quad (2.6)$$

The powers of $1 - Q$ cancel because $A = \sum_j a_j$. The classical limit $\Gamma_Q(u) \rightarrow \Gamma(u)$ for positive u proves $F_h(x) \rightarrow F_0(x)$ for each fixed $x > 0$ [7].

2.1 A finite Bernoulli expansion

Write $b = 2\pi$ and $b_\ell = b\ell$. The positive multinomial weight is

$$W(s) = \sum_j e^{-a_j s} - e^{-As} > 0 \quad (s \geq 0). \quad (2.7)$$

It satisfies $W(s) \leq re^{-a_* s}$, and its leading large- s term is $m_* e^{-a_* s}$. Let

$$c_k = \frac{B_{2k}}{(2k)!}, \quad V_\nu(\tau) = \text{Li}_\nu(e^{-A\tau}) - \sum_j \text{Li}_\nu(e^{-a_j \tau}). \quad (2.8)$$

For $h > 0$, define

$$\Phi_h(x) = \kappa h x^2 - d \log \mathcal{P}(h) + h^{-1} V_2(hx) - \frac{1}{2} V_1(hx), \quad (2.9)$$

$$T_K(h, x) = \Phi_h(x) + \sum_{k=1}^K c_k h^{2k-1} V_{2-2k}(hx), \quad K \in \mathbb{N} \cup \{0\}. \quad (2.10)$$

The exact product in Φ_h is retained. In particular, exponentially small modular information in that product has not been discarded. At the endpoint use

$$H_a = A \log A - \sum_j a_j \log a_j, \quad (2.11)$$

$$\Phi_0(x) = H_a x - \frac{d}{2} \log(2\pi x) + \frac{1}{2} \log \frac{A}{\prod_j a_j}, \quad (2.12)$$

$$T_K(0, x) = \Phi_0(x) + \sum_{k=1}^K \frac{B_{2k}}{2k(2k-1)} \frac{A^{1-2k} - \sum_j a_j^{1-2k}}{x^{2k-1}}. \quad (2.13)$$

These are finite sums even when K depends on x .

It is convenient to write both lattice and continuous formulas using a positive measure:

$$\int f(t) d\nu_h(t) = \begin{cases} h \sum f(hm), & h > 0, \\ \int_0^\infty f(t) dt, & h = 0. \end{cases} \quad (2.14)$$

Proposition 2.1 (Exact signed remainder). *For $x > 0$, $h \geq 0$ and $K \geq 0$,*

$$F_h(x) - T_K(h, x) = (-1)^{K+1} \mathcal{E}_0(2K; h, x), \quad (2.15)$$

where, more generally for $L \geq 0$ and $p \geq 0$,

$$\mathcal{E}_L(p; h, x) = 2 \int_0^\infty W(xt) \sum_{\ell=L+1}^\infty \frac{(t/b_\ell)^p}{b_\ell^2 + t^2} d\nu_h(t) > 0. \quad (2.16)$$

All displayed integrals and sums converge absolutely. They can be repeatedly differentiated in p on compact subsets of $p > -1$.

Proof. The partial fractions of $\coth$ [5] give

$$\frac{1}{e^v - 1} = \frac{1}{v} - \frac{1}{2} + 2v \sum_{\ell \geq 1} \frac{1}{b_\ell^2 + v^2}.$$

Apply the finite identity

$$\frac{1}{b_\ell^2 + v^2} = \sum_{j=0}^{K-1} \frac{(-1)^j v^{2j}}{b_\ell^{2j+2}} + \frac{(-1)^K v^{2K}}{b_\ell^{2K}(b_\ell^2 + v^2)}. \quad (2.17)$$

The Bernoulli coefficient identity $c_k = 2(-1)^{k-1}\zeta(2k)/(2\pi)^{2k}$ identifies the finite terms. In (2.4), the remaining kernel is multiplied by $-W(mhx)/m$. Since $v = hm$, this is exactly (2.15) for $h > 0$.

For $h = 0$, Binet's integral [6] in the form

$$\log \Gamma(u + 1) = \left(u + \frac{1}{2}\right) \log u - u + \frac{1}{2} \log(2\pi) + \int_0^\infty \frac{e^{-ut}}{t} \left(\frac{1}{e^t - 1} - \frac{1}{t} + \frac{1}{2}\right) dt$$

produces the same expression after taking the gamma quotient and applying (2.17). Near zero, $t^p |\log t|^j$ is integrable for $p > -1$. At infinity the ray weight decays exponentially, and the pole sum is bounded by a constant multiple of $\sum \ell^{-p-2}$ on compact p ranges. These bounds, with logarithmic factors included, justify the assertions about convergence and differentiation. Positivity follows from (2.7). $\square$

2.2 Resolving finitely many poles exactly

Define

$$T_{K,L}(h, x) = T_K(h, x) + (-1)^{K+1} 2 \int W(xt) \sum_{\ell=1}^L \frac{(t/b_\ell)^{2K}}{b_\ell^2 + t^2} d\nu_h(t). \quad (2.18)$$

The sum is empty at $L = 0$. Proposition 2.1 immediately gives

$$F_h(x) - T_{K,L}(h, x) = (-1)^{K+1} \mathcal{E}_L(2K; h, x). \quad (2.19)$$

Resolving a pole here means evaluating its positive remainder component, not appending a guessed exponential coefficient. Put

$$\beta = b(L + 1), \quad C_\beta(z) = \frac{m_* \sqrt{2\pi}}{\beta^{3/2} \sqrt{z}}. \quad (2.20)$$

All subsequent theorems allow an arbitrary fixed L . The case $L = 0$ is the original Bernoulli truncation problem.

3 Gaussian lattice sums and the zero-mean tilt

For $\lambda > 0$, $\theta \in \mathbb{R}/\mathbb{Z}$ and $\eta \in \mathbb{R}$, let

$$u_k = \lambda(k - \theta), \quad (3.1)$$

$$\mathcal{J}_\lambda(\theta, \eta) = \frac{\lambda}{\sqrt{2\pi}} \sum_{k \in \mathbb{Z}} \exp(-u_k^2/2 + \eta u_k), \quad (3.2)$$

$$M_j(\lambda, \theta, \eta) = \frac{\lambda}{\sqrt{2\pi}} \sum_{k \in \mathbb{Z}} u_k^j \exp(-u_k^2/2 + \eta u_k). \quad (3.3)$$

Thus $M_j = \partial_\eta^j \mathcal{J}$. At $\lambda = 0$, replace the normalized sum by the normalized Gaussian integral on $\mathbb{R}$; in particular, $\mathcal{J}_0(\theta, \eta) = e^{\eta^2/2}$. The phase is irrelevant at that endpoint.

Let $\Theta_\lambda(\theta) = \mathcal{J}_\lambda(\theta, 0)$. Completing the square and applying Gaussian Poisson summation give

$$\mathcal{J}_\lambda(\theta, \eta) = e^{\eta^2/2} \Theta_\lambda(\theta + \eta/\lambda), \quad (3.4)$$

$$\Theta_\lambda(\theta) = \sum_{n \in \mathbb{Z}} e^{-2\pi^2 n^2 / \lambda^2} e^{2\pi i n \theta} = 1 + 2 \sum_{n \geq 1} e^{-2\pi^2 n^2 / \lambda^2} \cos(2\pi n \theta). \quad (3.5)$$

This is the standard Gaussian instance of the theta transformation [8]. It also follows directly by computing the Fourier coefficients of the periodic Gaussian sum. The real Gaussian sum is useful for large λ ; its Fourier form is useful for small λ .

Lemma 3.1 (Uniform Gaussian lattice control). *Fix $H, M < \infty$ and a nonnegative integer j . The moments in (3.3), and any fixed number of their η derivatives, are uniformly bounded for*

$$0 \leq \lambda \leq H, \quad \theta \in \mathbb{R}/\mathbb{Z}, \quad |\eta| \leq M.$$

The normalized sums of $(1 + |u|^j) e^{-cu^2 + M|u|}$ are also bounded for any fixed $c > 0$. They have the corresponding integral limit uniformly in θ as $\lambda \downarrow 0$.

Proof. Partition the real line into unit intervals. An interval contains at most $1 + 1/\lambda$ lattice points, so its normalized counting mass is at most $1 + H$. The suprema of the Gaussian envelope on these intervals form a summable sequence. This proves domination, including polynomial factors. For the limit, first truncate to a compact interval. The shifted Riemann sums of a uniformly continuous function converge uniformly in their shifts. The dominated Gaussian tails are arbitrarily small uniformly in λ . Differentiation in η only adds polynomial factors. $\square$

Proposition 3.2 (The minimizing tilt). *For each $\lambda \geq 0$ and each phase θ , $\mathcal{J}_\lambda(\theta, \eta)$ has a unique minimizer $\eta_* = \eta_*(\lambda, \theta)$, characterized by*

$$M_1(\lambda, \theta, \eta_*) = 0. \quad (3.6)$$

Define

$$\mathcal{M}_\lambda(\theta) = \mathcal{J}_\lambda(\theta, \eta_*). \quad (3.7)$$

On every bounded interval $0 \leq \lambda \leq H$, η_ is bounded, $\mathcal{M}_\lambda$ is bounded away from zero, and $M_2(\lambda, \theta, \eta_*)$ is bounded away from zero, uniformly in θ . These functions are continuous with their endpoint values. At $\lambda = 0$, $\eta_* = 0$ and $\mathcal{M}_0 = 1$.*

Proof. The second η derivative is $M_2 > 0$: the lattice has points other than zero, all with positive weights. The integral case has the same property. There is positive mass on both positive and negative half-lines, so the function tends to infinity as $\eta \rightarrow \pm\infty$. This gives existence, strict uniqueness, and (3.6).

For uniformity, choose positive and negative compact intervals separated from zero, each long enough to contain several points of every lattice with $\lambda \leq H$. Their normalized Gaussian masses have positive lower bounds. The contributions from the interval of the same sign as η grow exponentially as $|\eta|$ grows; the opposite-sign derivative contribution is uniformly bounded and then decays. Hence the derivative has opposite signs at $\eta = \pm M_H$ for some fixed M_H . Lemma 3.1 and the implicit function theorem now give continuity of the zero and of the moments on the compact parameter set. Strict positivity and compactness give both lower bounds. The limiting statements also follow from (3.5) and (3.4). $\square$

The condition (3.6) has a simple interpretation: the optimal order reweights the lattice so that its weighted displacement from the continuous saddle is zero. At phases 0 and $1/2$, symmetry makes the optimal tilt zero. At a general phase it need not vanish. For example, the included numerical calculation at $(\lambda, \theta) = (4, 1/4)$ gives

$$\eta_* \approx 0.725347327957, \quad \mathcal{M}_4(1/4) \approx 0.624809319319. \quad (3.8)$$

These decimal values are diagnostics; their defining equations are exact.

4 The all-order critical-mesh theorem

Let L and the ray be fixed. For $h > 0$, introduce

$$\lambda = h\sqrt{z/\beta}, \quad \theta = \{\beta/h\}, \quad \varepsilon = (\beta z)^{-1/2}, \quad p = \beta z + \eta\sqrt{\beta z}. \quad (4.1)$$

At $h = 0$, set $\lambda = 0$ and any phase θ . The saddle coordinate is $u = (t - \beta)\sqrt{z/\beta}$, so $t = \beta(1 + \varepsilon u)$. For a lattice point $t = hm$, its u coordinate lies in $\lambda(\mathbb{Z} - \theta)$.

Define polynomials $P_j(u, \eta)$ by the formal expansion

$$\begin{aligned} \sum_{j \geq 0} P_j(u, \eta) \varepsilon^j &= \frac{2}{1 + (1 + \varepsilon u)^2} \\ &\times \exp \left\{ \frac{\log(1 + \varepsilon u) - \varepsilon u + (\varepsilon u)^2/2}{\varepsilon^2} + \eta \left(\frac{\log(1 + \varepsilon u)}{\varepsilon} - u \right) \right\}. \end{aligned} \quad (4.2)$$

The apparent singularities in ε are removable as formal series. Each P_j is a finite polynomial. In particular,

$$\begin{aligned} P_0 &= 1, \\ P_1 &= \frac{u^3}{3} - \frac{\eta u^2}{2} - u, \end{aligned} \quad (4.3)$$

$$P_2 = \frac{u^2}{2} + \frac{5\eta u^3}{6} + \left(\frac{\eta^2}{8} - \frac{7}{12} \right) u^4 - \frac{\eta u^5}{6} + \frac{u^6}{18}. \quad (4.4)$$

Let

$$\mathcal{J}_{j,\lambda}(\theta, \eta) = \frac{\lambda}{\sqrt{2\pi}} \sum_{k \in \mathbb{Z}} e^{-u_k^2/2 + \eta u_k} P_j(u_k, \eta), \quad (4.5)$$

with its Gaussian integral interpretation at $\lambda = 0$. Thus $\mathcal{J}_{0,\lambda} = \mathcal{J}_\lambda$.

Theorem 4.1 (Uniform theta expansion). *Fix $L \geq 0$, the positive ray $\mathbf{a}$, $H, M < \infty$, and integers $J, q \geq 0$. As $z \rightarrow \infty$,*

$$\frac{\mathcal{E}_L(p; h, x)}{C_\beta(z)e^{-\beta z}} = \sum_{j=0}^J \varepsilon^j \mathcal{J}_{j,\lambda}(\theta, \eta) + O_{\mathbf{a}, L, H, M, J, q}(\varepsilon^{J+1}), \quad (4.6)$$

uniformly for $h \geq 0$ with $0 \leq \lambda \leq H$ and $|\eta| \leq M$. The same estimate holds after any number at most q of η derivatives, where h, x are held fixed while p varies as in (4.1).

4.1 Isolating the first pole and the smallest ray

We give the estimates behind the uniform assertion rather than treating the moving lattice as an informal Riemann sum.

Lemma 4.2 (Unimodal sum bound). *If $f \geq 0$ is integrable and unimodal on $(0, \infty)$, then*

$$\int f \, d\nu_h \leq \int_0^\infty f(t) \, dt + h \sup f, \quad h > 0. \quad (4.7)$$

For $f(t) = t^p e^{-zt}$ with $p \geq 0$, this gives

$$\int f \, d\nu_h \leq \frac{\Gamma(p+1)}{z^{p+1}} + h \left(\frac{p}{ez} \right)^p, \quad (4.8)$$

where the last power equals 1 when $p = 0$.

Proof. On the increasing side, compare each lattice value with the integral on the interval to its right. On the decreasing side, use the interval to its left. These intervals are disjoint except for the single interval straddling the maximum. Accounting for that one value costs at most $h \sup f$. This proves (4.7); the gamma integral and the maximum at p/z give (4.8). $\square$

Lemma 4.3 (Exponential separation). *In the regime of Theorem 4.1, there is $c > 0$ such that*

$$\mathcal{E}_L(p; h, x) = 2m_* \int \frac{(t/\beta)^p e^{-zt}}{\beta^2 + t^2} \, d\nu_h(t) + O(C_\beta(z) e^{-\beta z - cz}). \quad (4.9)$$

The assertion remains valid after any fixed number of η derivatives, possibly with a smaller c .

Proof. For a later pole $b_\ell > \beta$, its integrand divided by the first-pole integrand is at most $(\beta/b_\ell)^p$. For $p = \beta z + O(\sqrt{z})$,

$$\sum_{\ell \geq L+2} (\beta/b_\ell)^p \leq \left(\frac{L+1}{L+2} \right)^p \left(1 + \frac{L+2}{p-1} \right),$$

which is exponentially small. The bound follows by comparing the decreasing power tail with an integral.

For a ray parameter $a > a_*$, set $\alpha = a/a_* > 1$. Its first-pole contribution has the upper bound

$$2\beta^{-p-2} \int t^p e^{-\alpha z t} \, d\nu_h(t).$$

By (4.8) and Stirling's formula, its ratio to $C_\beta(z) e^{-\beta z}$ is bounded by a polynomial in z times $\exp(-\beta z \log \alpha + O(\sqrt{z}))$. The same estimate applies to the negative numerator term with $a = A$. There are finitely many such parameters. For the smallest ray, (4.8) gives the required order upper bound on the first-pole contribution because $h = O(z^{-1/2})$. The claimed separation follows by decreasing c to absorb polynomial factors.

A η derivative multiplies an integrand by $\sqrt{\beta z} \log(t/b_\ell)$. Any fixed power of this factor is handled by splitting t into $(0, \beta/2)$, $[\beta/2, 2\beta]$, and $(2\beta, \infty)$ and using the same exponential moments. Equivalently, bound logarithmic powers by a small positive or negative power of t and ℓ ; such fixed changes do not alter a strict exponential gap. $\square$

4.2 Proof of Theorem 4.1

By Lemma 4.3, only the first term in (4.9) needs expansion. Substitution of $t = \beta(1 + \varepsilon u)$ gives its normalized form exactly as

$$\frac{\lambda}{\sqrt{2\pi}} \sum_{u \in \lambda(\mathbb{Z} - \theta), u > -1/\varepsilon} e^{-u^2/2 + \eta u} Q_\varepsilon(u, \eta), \quad (4.10)$$

where Q_ε is the right side of (4.2). At $h = 0$ this is the corresponding integral. The normalization uses $h\sqrt{z/\beta} = \lambda$ and is therefore valid without a limiting approximation of the lattice.

First restrict to $|u| \leq \varepsilon^{-1/12}$. Taylor's theorem applied to the logarithm, the exponential, and the rational prefactor shows, for every fixed J , that

$$\left| Q_\varepsilon - \sum_{j=0}^J \varepsilon^j P_j \right| \leq C \varepsilon^{J+1} (1 + |u|^{D_J}) e^{c_0 \varepsilon |u|^3 + c_1 \varepsilon u^2}. \quad (4.11)$$

Here D_J is a finite integer; the constants can be chosen uniformly for bounded η . The same form holds for finitely many η derivatives. On the chosen region the extra exponential is bounded by a Gaussian factor that leaves, for sufficiently small ε , a majorant $C(1 + |u|^{D_J}) e^{-u^2/4 + M|u|}$. Lemma 3.1 therefore makes the summed or integrated error $O(\varepsilon^{J+1})$.

For completeness, the complement is negligible uniformly in the mesh. On $t/\beta \in [1/2, 2]$, the function $\log(t/\beta) - t/\beta + 1$ is bounded above by $-c(t/\beta - 1)^2$. The tilt contributes at most $C|u|$ to the exponent. Thus the part with $|u| > \varepsilon^{-1/12}$ has a Gaussian tail $O(\exp(-c\varepsilon^{-1/6}))$. On $t/\beta < 1/2$ or $t/\beta > 2$, the same strict concavity gives an exponential gap in z ; the far right tail is bounded by an exponential moment, and the left tail by the integrable power t^p . Bound their lattice sums using (4.7) on each monotone segment. The resulting estimates are exponentially small after normalization. Finally, extending the polynomial Gaussian sums to the artificial points $u \leq -1/\varepsilon$ changes them by an exponentially small amount. These estimates prove (4.6), including the asserted derivative control. $\square$

Corollary 4.4 (What the conventional order rule actually gives). *If $p = 2K = \beta z + \sigma$ with bounded σ , then, uniformly for $0 \leq \lambda \leq H$,*

$$\mathcal{E}_L(2K; h, x) = C_\beta(z) e^{-\beta z} \left(\Theta_\lambda(\theta) + O(z^{-1/2}) \right). \quad (4.12)$$

The first correction is the normalized Gaussian lattice average of $u^3/3 + (\sigma - 1)u$, multiplied by ε .

Proof. Set $\eta = \sigma\varepsilon$ in Theorem 4.1 and expand once more in η . The additional term is σM_1 . $\square$

The theorem identifies the transition asked for in the preceding draft, but (4.12) alone is not an optimal-truncation theorem. That extra minimization is the next step.

5 Global optimal truncation and the phase-dependent order shift

Lemma 5.1 (Strict convexity and a local certificate for a global minimum). *For fixed h, x, L , the function $p \mapsto \mathcal{E}_L(p; h, x)$ is strictly log-convex on $p > -1$. It tends to infinity as $p \rightarrow \infty$. In particular, its sampled ratios*

$$\frac{\mathcal{E}_L(2K + 2; h, x)}{\mathcal{E}_L(2K; h, x)} \quad (5.1)$$

are strictly increasing in K . An integer $K \geq 1$ whose value is no larger than either neighboring value is a global minimizer. There are at most two integer minimizers, and two minimizers must be adjacent.

Proof. Expression (2.16) is a positive mixture of exponentials $\exp(p \log(t/b_\ell))$. Normalize it to a probability measure on pairs (t, ℓ) . Twice differentiating its logarithm gives

$$\frac{d^2}{dp^2} \log \mathcal{E}_L(p; h, x) = \text{Var}_p(\log(t/b_\ell)) > 0.$$

The variance is strict because the support contains distinct ratios t/b_ℓ . To see growth at infinity, retain one interval of $t > \beta$ at $h = 0$, or one lattice point $hm > \beta$ at $h > 0$, and retain the first pole. Its positive contribution grows exponentially in p . The difference of a strictly convex function between consecutive points of a fixed-step grid is strictly increasing. Apply this to $\log \mathcal{E}_L$. $\square$

All moments in the next theorem are evaluated at $(\lambda, \theta, \eta_*(\lambda, \theta))$. Define

$$s_*(\lambda, \theta) = \frac{3}{2} - \frac{M_4}{3M_2} + \frac{\eta_* M_3}{2M_2}, \quad (5.2)$$

$$B_*(\lambda, \theta) = \frac{M_3}{3} - \frac{\eta_* M_2}{2}. \quad (5.3)$$

Both are finite and bounded on a bounded mesh interval.

Theorem 5.2 (The actual global minimum). *Fix the ray, L , and $H < \infty$. Uniformly for $0 \leq h\sqrt{z/\beta} \leq H$, the continuous minimizer p_c of $\mathcal{E}_L(p; h, x)$ on $p \geq 0$ is interior for large z and satisfies*

$$p_c = \beta z + \eta_* \sqrt{\beta z} + s_* + O(z^{-1/2}). \quad (5.4)$$

Every integer minimizer over $K \geq 0$ satisfies

$$2K_{\min} = \beta z + \eta_* \sqrt{\beta z} + O(1). \quad (5.5)$$

More precisely, it is one of the two even integers bracketing p_c . The minimum error obeys

$$\min_{K \geq 0} \mathcal{E}_L(2K; h, x) = C_\beta(z) e^{-\beta z} \left(\mathcal{M}_\lambda(\theta) + \varepsilon B_*(\lambda, \theta) + O(z^{-1}) \right). \quad (5.6)$$

In particular, the leading formula is a uniform relative asymptotic statement, since $\mathcal{M}_\lambda$ is bounded away from zero for $\lambda \leq H$.

Proof. Choose M large enough that the derivative of $\mathcal{J}_\lambda$ has opposite signs at $\eta = \pm M$, uniformly over the parameter set, as in Proposition 3.2. The derivative form of Theorem 4.1 then gives the same signs for $\mathcal{E}_L$ at $p = \beta z \pm M\sqrt{\beta z}$. Strict log-convexity supplies a unique critical point between them, which is the global minimum. This also excludes the boundary $p = 0$ for large z ; no assumption about the likely location of the global minimizer has been made.

Write $\eta_c = (p_c - \beta z)\varepsilon$. Differentiating the expansion twice and using the uniform positive lower bound for M_2 gives

$$\eta_c = \eta_* - \varepsilon \frac{\partial_\eta \mathcal{J}_{1,\lambda}(\theta, \eta_*)}{M_2} + O(\varepsilon^2).$$

The coefficient from (4.3) is

$$\mathcal{J}_1 = M_3/3 - \eta M_2/2 - M_1.$$

Consequently

$$\partial_\eta \mathcal{J}_1 = M_4/3 - \eta M_3/2 - 3M_2/2.$$

Substitution proves (5.4) and (5.2).

A strictly convex function decreases before its continuous minimum and increases after it. Hence its minimum on the even integers is attained among the two bracketing even integers, and each is within 2 of p_c . This proves (5.5). Such a displacement changes η by $O(\varepsilon)$. Because the first derivative of the leading profile vanishes at η_* , evaluating at the integer minimizer changes its leading value only by $O(\varepsilon^2)$. The first correction at η_* is $\mathcal{J}_1 = B_*$ because $M_1 = 0$. This proves (5.6). $\square$

5.1 An effective order-selection procedure

The leading tilt requires only a one-dimensional strictly monotone root search for $M_1 = 0$. With that tilt, form

$$p_{\text{pred}} = \beta z + \eta_* \sqrt{\beta z} + s_*. \quad (5.7)$$

For sufficiently large z , evaluating the three integer orders centered at the nearest integer to $p_{\text{pred}}/2$ includes a true minimizer. At any finite z , the neighboring-ratio certificate of Lemma 5.1, together with the finite error bounds in Section 9, certifies global minimality whenever the resulting comparison intervals are disjoint. This separates a very useful asymptotic guess from a logically sufficient finite certificate.

Theorem 5.2 also gives a precise sense in which the ordinary rule can be suboptimal:

$$\frac{\text{error at } 2K = \beta z + O(1)}{\text{minimum error}} \longrightarrow \frac{\Theta_\lambda(\theta)}{\mathcal{M}_\lambda(\theta)} \quad (5.8)$$

along any sequence with fixed limiting mesh and phase. The ratio is at least one, and is strictly larger than one whenever $\eta_* \neq 0$.

5.2 A second-order classical endpoint law

At $h = 0$, all odd Gaussian moments vanish and a more precise formula is particularly simple.

Corollary 5.3 (Integer-order oscillation at the continuous endpoint). *Let $h = 0$ and keep the ray and L fixed. Put $d_z = \text{dist}(\beta z + \frac{1}{2}, 2\mathbb{Z})$. Then*

$$\min_{K \geq 0} \mathcal{E}_L(2K; 0, x) = C_\beta(z) e^{-\beta z} \left[1 + \frac{d_z^2/2 - 13/24}{\beta z} + O(z^{-2}) \right]. \quad (5.9)$$

The continuous minimizing order is $p_c = \beta z + \frac{1}{2} + O(z^{-1})$.

Proof. For a bounded offset $p = \beta z + \sigma$, expansion of the exact scaled integral gives a zero odd correction and second coefficient

$$\frac{\sigma^2 - 3\sigma + 1}{2} u^2 + \frac{4\sigma - 7}{12} u^4 + \frac{u^6}{18}.$$

The Gaussian moments 1, 3, 15 yield

$$\frac{\mathcal{E}_L(\beta z + \sigma; 0, x)}{C_\beta(z) e^{-\beta z}} = 1 + \frac{\sigma^2/2 - \sigma/2 - 5/12}{\beta z} + O(z^{-2}). \quad (5.10)$$

The next odd power also integrates to zero. The polynomial is $\frac{1}{2}(\sigma - \frac{1}{2})^2 - 13/24$. Theorem 5.2 restricts integer minimizers to a uniformly bounded offset, so minimizing this expression on the even-order grid proves (5.9); a possible tie changes only the stated $O(z^{-2})$ term. Differentiating the same endpoint expansion proves the assertion for p_c . $\square$

6 The fixed-mesh arithmetic action

The critical-mesh theorem is uniform for bounded λ , not for all h . When h is fixed, the saddle width shrinks while the lattice spacing does not. A different minimization becomes visible. The exact positive remainder makes it possible to solve this problem without extrapolating Theorem 4.1 outside its hypotheses.

Fix $0 < h < \beta$ and assume $\beta/h \notin \mathbb{Z}$. Let

$$m = \lfloor \beta/h \rfloor, \quad t_- = hm, \quad t_+ = h(m+1), \quad (6.1)$$

$$d_- = \log(t_-/\beta) < 0, \quad d_+ = \log(t_+/\beta) > 0, \quad (6.2)$$

$$\rho_h = \frac{t_+ - t_-}{d_+ - d_-}, \quad I_h = t_- - \rho_h d_- = t_+ - \rho_h d_+, \quad (6.3)$$

$$c_{\pm} = \frac{2m_*h}{\beta^2 + t_{\pm}^2}, \quad C_h(s) = c_- e^{sd_-} + c_+ e^{sd_+}, \quad (6.4)$$

$$s_h = \frac{1}{d_+ - d_-} \log \frac{-c_- d_-}{c_+ d_+}. \quad (6.5)$$

The number ρ_h is the logarithmic mean of t_- and t_+ , so it lies strictly between them. The real number s_h is the unique minimizer of C_h .

Theorem 6.1 (Optimal error at a fixed nonresonant mesh). *Fix the ray, L , and h as above. For some $\gamma > 0$, the continuous minimizer and the minimum over integer orders satisfy*

$$p_c = \rho_h z + s_h + O(e^{-\gamma z}), \quad (6.6)$$

$$\min_{K \geq 0} \mathcal{E}_L(2K; h, x) = e^{-I_h z} \left[\min_{K \in \mathbb{Z}} C_h(2K - \rho_h z) + O(e^{-\gamma z}) \right]. \quad (6.7)$$

The minimum in brackets is positive, bounded, and periodic in $\rho_h z$ with period 2. Moreover,

$$I_h > \beta. \quad (6.8)$$

The estimates can be made uniform for h in a compact subset of one interval between consecutive resonance values β/n .

Proof. At $p = \rho_h z + s$, the two first-pole, smallest-ray contributions at t_- and t_+ equal

$$e^{-I_h z} (c_- e^{sd_-} + c_+ e^{sd_+}).$$

For every other lattice point, use the phase

$$\phi(t) = \rho_h \log(t/\beta) - t.$$

It is strictly concave, increases before $t = \rho_h$, and decreases after it. Since $t_- < \rho_h < t_+$, its largest lattice values are precisely the equal values at t_- and t_+ . The gap to the remaining fixed lattice is strictly positive. For large t , the linear term $-t$ controls the infinite tail. Thus all other first-pole lattice contributions are $O(e^{-(I_h + \gamma_1)z})$, uniformly for bounded s , with the same estimate for any fixed number of s derivatives.

Later poles are bounded relative to the first-pole sum by $\sum_{\ell \geq L+2} (\beta/b_\ell)^p$ and hence are exponentially smaller. For a ray parameter larger than a_* , each lattice contribution gains at least $\exp(-(a/a_* - 1)zh)$. This also covers the negative numerator term by an absolute bound. Therefore, on each bounded s interval,

$$e^{I_h z} \mathcal{E}_L(\rho_h z + s; h, x) = C_h(s) + O(e^{-\gamma z}), \quad (6.9)$$

with derivative control.

The derivative of C_h changes sign at s_h . Choose fixed points on either side; the same sign change holds for the exact remainder for large z . Lemma 5.1 brackets its unique global minimizer there. Since $C_h''(s_h) > 0$, the differentiated expansion proves (6.6). Integer minimizers are the even integers bracketing p_c . Their offsets $2K - \rho_h z$ are bounded, so (6.9) proves (6.7). The minimum of C_h on a shifted step-two grid is continuous, positive, and periodic in the grid shift, giving the prefactor assertions.

Finally, consider the strictly convex graph $t = \beta e^d$. Its chord between d_- and d_+ has equation $t = I_h + \rho_h d$. Because 0 is strictly between the two endpoints, the chord lies strictly above the graph there: $I_h > \beta$. Compactness away from resonances preserves all the positive gaps and derivative lower bounds used above. $\square$

Example 6.2 (An action larger than 2π). For the ordinary remainder, $\beta = 2\pi$. At $h = 4$, the straddling points are $t_- = 4$ and $t_+ = 8$, giving

$$\rho_h = \frac{4}{\log 2} \approx 5.77078016356, \quad I_h = 4 - \frac{4}{\log 2} \log \frac{2}{\pi} \approx 6.60598451789.$$

Thus the minimum is of order $e^{-6.6059845z}$ times a bounded positive periodic prefactor, not merely $e^{-2\pi z}$ times a constant. This is an improvement for the same family of ordinary Bernoulli truncations, not a change of numerical representation.

Proposition 6.3 (Resonant meshes remain an obstruction). *If $h = \beta/m_0$ for an integer $m_0 \geq 1$, then*

$$\min_{K \geq 0} \mathcal{E}_L(2K; h, x) \sim \frac{m_* h}{\beta^2} e^{-\beta z}. \quad (6.10)$$

Proof. For every p , the first-pole contribution at the lattice point $t = \beta$ is $hW(\beta x)/\beta^2$. It is independent of p and asymptotic to the right-hand side of (6.10). Positivity gives a lower bound for every order. For an upper bound choose any even $p = \beta z + O(1)$. The phase $\beta \log(t/\beta) - t$ has its strict maximum at the lattice point β ; all other fixed lattice points have an exponential gap. Later poles and larger rays are exponentially smaller by the arguments used in Theorem 6.1. This proves a matching upper bound. $\square$

The resonant leading minimum does not select a universal minimizing order: a whole macroscopic interval of orders can share the same leading irreducible contribution, with smaller terms deciding the optimum. We do not assign an unproved order law at exact resonance. Similarly, Theorem 6.1 is not a claim for $h > \beta$, where there is no positive lattice point below the first unresolved pole.

7 Matching the theta law to the two-point law

A bounded- λ theorem cannot simply be declared uniform for $\lambda \rightarrow \infty$. We first analyze the theta profile itself, then prove an overlap statement directly from the exact remainder.

Proposition 7.1 (Large-mesh theta minimization). *Fix $0 < \delta < 1/2$. Uniformly for $\delta \leq \theta \leq 1 - \delta$, as $\lambda \rightarrow \infty$,*

$$\eta_*(\lambda, \theta) = \lambda(\tfrac{1}{2} - \theta) + \lambda^{-1} \log \frac{\theta}{1 - \theta} + O_\delta(\lambda^{-1} e^{-\lambda^2}), \quad (7.1)$$

$$\mathcal{M}_\lambda(\theta) = \frac{\lambda}{\sqrt{2\pi}} \exp \left[-\frac{\lambda^2}{2} \theta(1 - \theta) + H(\theta) \right] (1 + O_\delta(e^{-\lambda^2})), \quad (7.2)$$

$$H(\theta) = -\theta \log \theta - (1 - \theta) \log(1 - \theta).$$

Proof. Retain only $u_- = -\lambda\theta$ and $u_+ = \lambda(1 - \theta)$. At zero weighted mean their weight ratio is $\theta/(1 - \theta)$. Solving for the tilt gives the first two terms of (7.1). At that tilt their total weight is

$$\exp[-\lambda^2\theta(1 - \theta)/2]\theta^{-\theta}(1 - \theta)^{-(1-\theta)}.$$

The next point to either side is smaller than the corresponding retained point by $O_\delta(e^{-\lambda^2})$. The remaining points have still larger quadratic gaps and form a dominated geometric tail. Their first moment has a relative factor $O(\lambda e^{-\lambda^2})$, whereas the two-point second moment is bounded below by a positive constant times λ^2 times the two-point mass. The correction to the optimizing tilt is therefore $O_\delta(\lambda^{-1}e^{-\lambda^2})$. The same bounds give the stated relative error for the minimum. $\square$

Theorem 7.2 (A genuine mesoscopic overlap). *Fix the ray, L , and $0 < \delta < 1/2$. Suppose that, as $z \rightarrow \infty$,*

$$h \rightarrow 0, \quad zh^2 \rightarrow \infty, \quad zh^3 \rightarrow 0, \quad \delta \leq \{\beta/h\} \leq 1 - \delta. \quad (7.3)$$

Writing $\theta = \{\beta/h\}$, the actual integer-order minimum satisfies

$$\min_{K \geq 0} \mathcal{E}_L(2K; h, x) \sim \frac{m_* h}{\beta^2} \exp \left[-\beta z - \frac{zh^2}{2\beta} \theta(1 - \theta) + H(\theta) \right]. \quad (7.4)$$

This is exactly the large- λ expression from (7.2) multiplied by $C_\beta(z)e^{-\beta z}$.

Proof. Use $t_- = \beta - h\theta$ and $t_+ = \beta + h(1 - \theta)$, and the exact quantities (6.3)–(6.5). Uniform Taylor expansion for $\theta \in [\delta, 1 - \delta]$ gives

$$\rho_h = \beta + h(\tfrac{1}{2} - \theta) + O(h^2), \quad (7.5)$$

$$I_h = \beta + \frac{h^2}{2\beta} \theta(1 - \theta) + O(h^3), \quad (7.6)$$

$$d_- = -h\theta/\beta + O(h^2), \quad d_+ = h(1 - \theta)/\beta + O(h^2),$$

$$c_\pm = m_* h \beta^{-2} (1 + O(h)), \quad s_h = \frac{\beta}{h} \log \frac{\theta}{1 - \theta} + O(1). \quad (7.7)$$

At $p = \rho_h z + s_h$, the two retained terms consequently have total

$$e^{-I_h z} \frac{m_* h}{\beta^2} e^{H(\theta)} (1 + O(h)). \quad (7.8)$$

Moving p by at most one to an even integer changes this expression by $1 + O(h^2)$ because its first derivative vanishes and both logarithmic slopes are $O(h)$.

We justify using this pair as h varies. For lattice points a bounded number of steps away, strict concavity of $\rho_h \log(t/\beta) - t$ gives a phase gap at least a positive constant times h^2 to the next omitted point. Further local points have quadratic step gaps. The term $s_h \log(t/\beta)$ changes neighboring logarithmic weights by bounded amounts, since $s_h = O(1/h)$. Thus the total omitted local contribution relative to the pair is bounded by $C_\delta \sum_{j \geq 1} \exp(-c_\delta z h^2 j^2 + C_\delta j)$, which tends to zero. Outside a fixed neighborhood of β , the phase gap is of order z , whereas the tilt is $O(1/h) = o(z)$. Moment bounds handle the infinite tail. Later poles and nonminimal rays remain exponentially smaller.

For a lower bound, use positivity and note uniformly near the two retained points that $W(xt) = m_* e^{-zt}(1 + O(e^{-cz}))$. The minimum of the two-point contribution over real p bounds the exact integer-order minimum from below up to this relative error. Evaluating at the even integer just chosen gives the matching upper bound. This establishes (7.8) for the exact minimum. Finally, $zh^3 \rightarrow 0$ allows (7.6) to be exponentiated with relative error tending to zero, yielding (7.4). $\square$

The overlap is nonempty: $h = z^{-\alpha}$ with $1/3 < \alpha < 1/2$, restricted to phases away from resonance, is an example. Near resonance, a different phase boundary layer is required; it is listed as a further research question rather than hidden in the word “uniform.”

8 An exact hierarchy beyond the first exponential scale

Theorem 8.1 (Reoptimized pole resolution). *For every fixed $L \geq 0$, the approximants $T_{K,L}$ have the exact signed error (2.19). Their critical-mesh optimal error and optimal orders are given by Theorems 4.1 and 5.2 with*

$$\beta = 2\pi(L + 1). \quad (8.1)$$

In particular, at $h = 0$,

$$\min_{K \geq 0} |F_0(x) - T_{K,L}(0, x)| \sim \frac{m_* \sqrt{2\pi}}{[2\pi(L + 1)]^{3/2} \sqrt{z}} e^{-2\pi(L+1)z}. \quad (8.2)$$

For bounded $h\sqrt{z/[2\pi(L + 1)]}$ the same formula has the optimized theta factor from (5.6). Fixed-mesh and resonance formulas also apply with this new β .

Proof. The exact construction removes the first L terms of the positive pole sum and nothing else. The first remaining denominator is $\beta^2 + t^2$, and its smallest-ray exponential is e^{-zt} . These are precisely the hypotheses used throughout Sections 4–7. No further limiting interchange is involved. The endpoint formula is Corollary 5.3 at leading order. $\square$

The resolved component includes the entire weight W , not only its smallest-ray term. Thus no contribution from a larger ray is left behind at a resolved pole. This distinction matters when comparing the next action with the second-smallest ray parameter.

For $L = 1$, this replaces $e^{-2\pi z}$ by $e^{-4\pi z}$ in the endpoint and critical-mesh scales. The order must be reoptimized around $2K = 4\pi z$, not left near $2K = 2\pi z$. Removing a leading pole at the old order and then claiming the new least remainder would be incorrect. The new coefficient is a function or integral resolved from the exact kernel; it is not a conclusion about a convergent expansion in $e^{-2\pi z}$ with scalar coefficients.

There is also a numerical caution. At the higher order, T_K and the resolved correction can be much larger than their difference. The exact error theorem does not itself make subtractive evaluation well conditioned. A stable implementation should combine kernels before evaluation or work with sufficient guard precision. Section 9 controls truncation of positive sums; arithmetic roundoff still needs its own validated enclosure.

9 Finite evaluation and certified inverse errors

The asymptotic theorems locate the answer efficiently. To certify a specific numerical evaluation, it is preferable to bound the positive exact remainder directly rather than estimate an unspecified big- O constant.

For $p \geq 0$, set

$$Z_{p,L} = \sum_{\ell \geq L+1} b_\ell^{-p-2} = b^{-p-2} \zeta(p+2, L+1), \quad (9.1)$$

where the last notation is the Hurwitz zeta function. This quantity has elementary decreasing-series upper bounds, so it need not be treated as an oracle.

Theorem 9.1 (Finite positive-sum certificate). *Let $h > 0$, $p \geq 0$, $M \geq 1$, $J \geq L + 1$, and suppose $Mh \geq p/z$. Define the finite sum*

$$S_{M,J} = 2h \sum_{m=1}^M W(xhm) \sum_{\ell=L+1}^J \frac{(hm/b_\ell)^p}{b_\ell^2 + (hm)^2}. \quad (9.2)$$

Then

$$0 \leq \mathcal{E}_L(p; h, x) - S_{M,J} \leq U_m + U_\ell, \quad (9.3)$$

where

$$U_m = 2rZ_{p,L}z^{-p-1}\Gamma(p+1, zMh), \quad (9.4)$$

$$U_\ell = 2rZ_{p,J} \left[\frac{\Gamma(p+1)}{z^{p+1}} + h \left(\frac{p}{ez} \right)^p \right], \quad (9.5)$$

$$Z_{p,J} \leq b^{-p-2} \left[(J+1)^{-p-2} + \frac{(J+1)^{-p-1}}{p+1} \right]. \quad (9.6)$$

Here $\Gamma(s, y) = \int_y^\infty t^{s-1}e^{-t} dt$, and the peak power is 1 at $p = 0$.

Proof. Every omitted term is positive. In the part with $m > M$, bound $W(xt) \leq re^{-zt}$ and $(b_\ell^2 + t^2)^{-1} \leq b_\ell^{-2}$. The function $t^p e^{-zt}$ is decreasing for $t \geq p/z$, so

$$h \sum_{m>M} (hm)^p e^{-zhm} \leq \int_{Mh}^\infty t^p e^{-zt} dt = z^{-p-1} \Gamma(p+1, zMh).$$

This proves (9.4). The part with $\ell > J$, including all m , is bounded in the same way and then by Lemma 4.2, giving (9.5). Overcounting the intersection of these omitted regions is harmless. Finally, compare the decreasing pole-power tail, after its first term, with its integral to obtain (9.6). $\square$

For integer p , the incomplete gamma factor can be evaluated using the finite expression

$$\Gamma(p+1, y) = p!e^{-y} \sum_{j=0}^p \frac{y^j}{j!}. \quad (9.7)$$

All computations should be scaled logarithmically when p and z are large. At $h = 0$, the same proof bounds the continuous tail beyond a chosen $T \geq p/z$ by $2rZ_{p,L}z^{-p-1}\Gamma(p+1, zT)$. The remaining compact integral requires a separate validated quadrature bound.

These are analytic truncation bounds. To obtain machine-certified intervals, the finite sums, elementary functions, and quadrature must also be evaluated with outward-rounded interval arithmetic. The included mpmath implementation reports the analytic tail bounds using high precision, but does not claim to perform that additional step.

9.1 A uniform slope bound

The following conditioning result is rederived to keep the inverse certificate independent of the status of the preceding manuscript.

Lemma 9.2 (Uniform conditioning). *Put $x_0 = 1/a_*$ and*

$$c_a = A\psi(Ax_0 + 1) - \sum_j a_j \psi(a_j x_0 + 1) > 0. \quad (9.8)$$

For $h \geq 0$ and $x \geq x_0$,

$$F'_h(x) \geq F'_0(x) \geq c_a. \quad (9.9)$$

For the central ray $(1, 1)$, $x_0 = 1$ and $c_a = 1$.

Proof. For $h > 0$, differentiation of the absolutely convergent series gives

$$F'_h(x) = 2\kappa hx + h \sum_{m \geq 1} g_x(hm), \quad g_x(t) = \frac{\sum_j a_j e^{-a_j x t} - A e^{-A x t}}{e^t - 1}, \quad g_x(0) = 2\kappa x.$$

For $x \geq x_0$, this is a sum of positive decreasing functions

$$a_j e^{-(a_j x + 1)t} \frac{1 - e^{-(A - a_j)x t}}{1 - e^{-t}}.$$

Indeed $(A - a_j)x \geq 1$, and for $v \geq 1$ the logarithmic derivative of $(1 - e^{-vt})/(1 - e^{-t})$ is $v/(e^{vt} - 1) - 1/(e^t - 1) \leq 0$. The lower Riemann-sum bound consequently yields

$$h \sum_{m \geq 1} g_x(hm) \geq \int_0^\infty g_x(t) dt - h g_x(0).$$

The last term cancels the quadratic derivative. The remaining integral is $F'_0(x)$ by the integral representation of digamma differences [6].

For positivity and monotonicity, the trigamma integral gives

$$F''_0(x) = \int_0^\infty s e^{-xs} \left[\frac{1}{e^{s/A} - 1} - \sum_j \frac{1}{e^{s/a_j} - 1} \right] ds > 0.$$

To see the strict sign, the function $u \mapsto [u(e^{s/u} - 1)]^{-1}$ is strictly increasing. Therefore each term with $a_j < A$ is less than a_j/A times the first term in the brackets. Summing proves the sign. Since $F'_0(0) = 0$, $F'_0(x) > 0$ for $x > 0$, and (9.8)–(9.9) follow. For $(1, 1)$, $2\psi(3) - 2\psi(2) = 1$. $\square$

Theorem 9.3 (Certified inversion of a pole-resolved approximation). *Fix $h \geq 0$, let $y \geq F_h(x_0)$, and let x_* be the unique solution of $F_h(x_*) = y$. For any $s \geq x_0$, suppose an evaluated approximation $\tilde{T}$ satisfies*

$$|\tilde{T} - T_{K,L}(h, s)| \leq \delta_T, \quad \mathcal{E}_L(2K; h, s) \leq U.$$

Then

$$|x_* - s| \leq \frac{|\tilde{T} - y| + \delta_T + U}{c_a}. \quad (9.10)$$

For an exactly solved equation $T_{K,L}(h, s) = y$, the error is one-sided: $x_ \geq s$ when K is even, and $x_* \leq s$ when K is odd.*

Proof. The series formula shows that F_h is increasing on $(0, \infty)$, and it tends to infinity; the same is true at $h = 0$. Thus the stated solution exists uniquely on $[x_0, \infty)$. Its residual at s is bounded by the numerator of (9.10), using (2.19). The mean-value theorem and Lemma 9.2 give the conclusion. In the exactly solved case, the sign $(-1)^{K+1}$ determines the side of the root. $\square$

The derivative in this theorem holds h fixed. A path $h = h(x)$ used to state an asymptotic regime does not change that convention. Differentiating along such a path and reusing the same slope bound would be unjustified.

10 Reproducible calculations and numerical evidence

The package contains `code/verify.py`, CSV tables, and a JSON receipt. The script uses 75 decimal digits. It checks nine exact finite geometric identities, the polynomials P_1, P_2 , the quadratic action expansion, and the endpoint coefficient $-13/24$ symbolically. It also checks the signed forward-kernel identity for seven small truncation orders at a fixed positive (h, x) .

The main diagnostics compare the positive double remainder with the theta and two-point formulas. The main values are computed directly from finite positive sums. Omitted-sum bounds come from Theorem 9.1. Those bounds do not include floating-point rounding; this is why the receipt is labeled a diagnostic rather than a formal numerical proof. The comparisons at neighboring integer orders verify the expected local minimum with a large numerical separation. In exact interval arithmetic, Lemma 5.1 would turn the same comparisons into a global minimum certificate.

10.1 Critical-mesh experiments

For fixed λ, θ , choose an integer n and set

$$h = \frac{2\pi}{n + \theta}, \quad z = \frac{\lambda^2(n + \theta)^2}{2\pi}. \quad (10.1)$$

This makes the scaled mesh and phase exact by construction. The central ray $(1, 1)$ has $m_* = 2$ and $z = x$. Let K_0 denote the nearest integer to πz . In the following table, “gain” is the error at K_0 divided by the error at the observed minimum. “Leading relative error” compares the normalized minimum with $\mathcal{M}_\lambda(\theta)$.

λ	n	K_0	$K_{\min}$	leading rel. error	gain
4.0	8	544	556	0.0262	1.61802
4.0	16	2112	2136	0.0134	1.59767
4.0	32	8320	8367	0.00692	1.58729
6.0	8	1225	1257	0.0433	5.26040
6.0	16	4753	4816	0.0231	5.32586
6.0	32	18721	18848	0.0117	5.36548

The optimal shift grows linearly with n , hence like $\sqrt{z}$, as predicted by Theorem 5.2. The full CSV includes meshes $\lambda = 1, 2, 4, 6$, a symmetric phase $1/2$, and the first corrected error from (5.6). At the symmetric phase the leading tilt vanishes; that is an important control case rather than a universal feature.

10.2 Fixed-mesh experiments

For $h = 2\pi/2.25$ and $h = 4$, we compare $e^{I_h z} \min_K \mathcal{E}_0(2K; h, x)$ with the exact two-point-grid minimum in (6.7). The rapidly decreasing relative difference is consistent with an exponentially small omitted contribution.

h	z	I_h	$K_{\min}$	two-atom rel. error
2.79253	10	6.3962511	34	4.85e-05
2.79253	25	6.3962511	86	3.7e-10
2.79253	60	6.3962511	206	9.58e-23
4.00000	10	6.6059845	30	6.38e-08
4.00000	25	6.6059845	73	6.88e-19
4.00000	60	6.6059845	174	4.07e-44

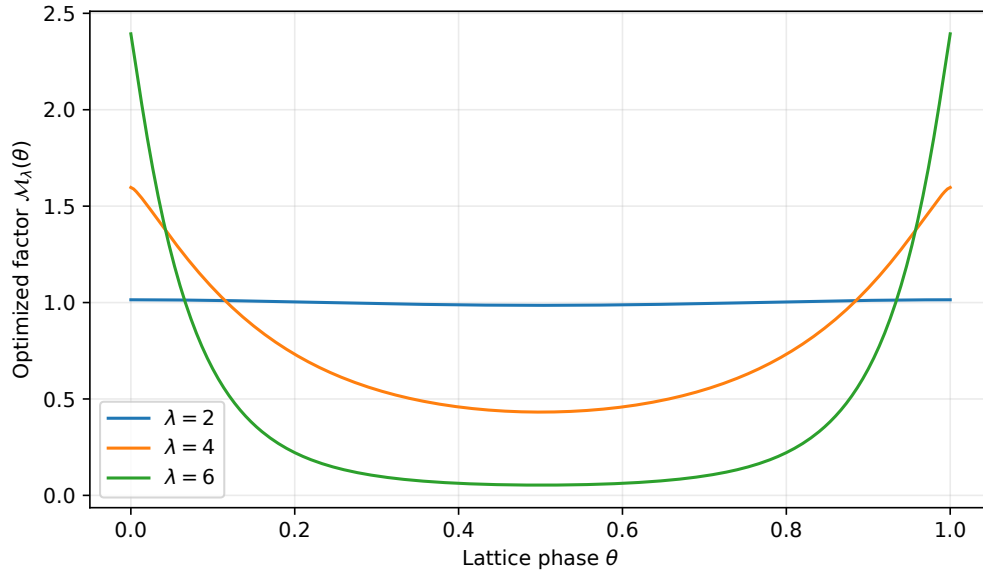


Figure 1: The optimized theta factor. The mesh and phase survive in the leading remainder; replacing the sum by its integral would give the constant function 1. This plot illustrates the exact profile definition, not a numerical proof of the asymptotic theorem.

The slight oscillation in the scaled minimum itself is retained in the CSV. It is described by the shifted step-two grid, and is not an error that should be averaged away.

10.3 Pole resolution and checks not claimed

Three additional cases resolve $L = 0, 1, 2$ poles while keeping the same scaled mesh and phase. They give the same normalized leading theta law, with the expected replacement of β by $2\pi(L + 1)$. The receipt records all 15 critical-mesh cases, six fixed-mesh cases, and three pole-resolution cases, with all assertions passing.

The experiments do not establish the all-order theorem, do not verify complex Stokes data, and do not prove a claim about an arbitrary analytic interpolation. Their purposes are to detect signs, normalization mistakes, order-rounding errors, and a failed asymptotic prediction. The mathematical justification is given in the preceding proofs.

11 Interpretation for the transseries program

11.1 Three distinct levels of asymptotic information

For this family, the following levels should not be conflated.

Formal coefficients. The Bernoulli–polylogarithm expressions determine every fixed-order coefficient. They do not by themselves determine the size of a remainder when the number of terms grows with x .

A specified near-least-term remainder. The rule $2K = \beta z + O(1)$ produces a theta factor $\Theta_\lambda(\theta)$. This gives the lattice-to-integral transition, but it is still the error of a specified rule.

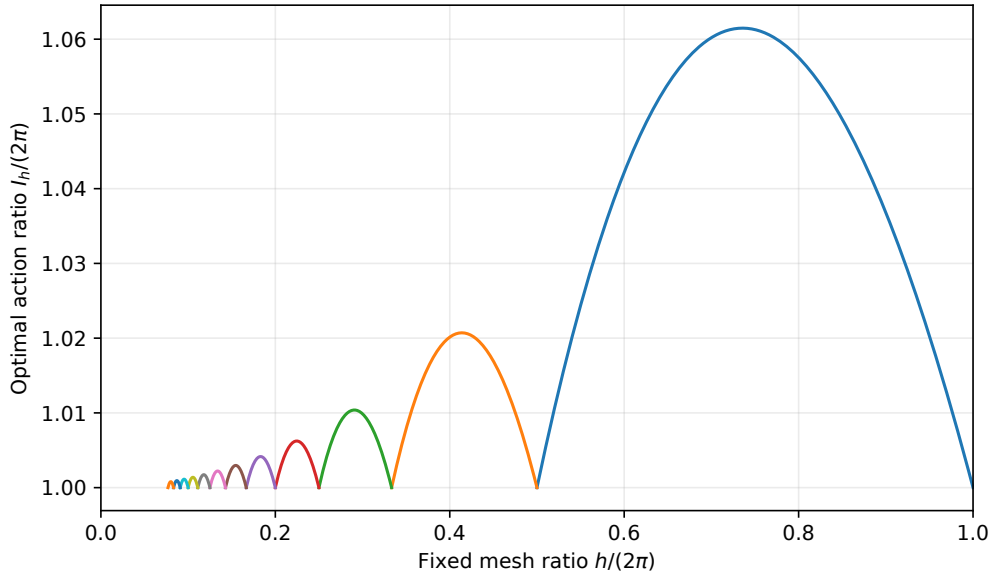


Figure 2: The fixed-mesh action $I_h/(2\pi)$ for the ordinary remainder. It is strictly greater than 1 between resonances and equals 1 at $h/(2\pi) = 1/n$. The theorem applies on the open nonresonant intervals; Proposition 6.3 treats their resonant endpoints.

The actual minimum and a refined representation. Minimization changes the leading factor to $\mathcal{M}_\lambda(\theta)$ and usually shifts the order by $\sqrt{z}$. At fixed nonresonant mesh, the action changes to I_h . Resolving exact pole components changes the representation itself and moves the first unresolved action to a larger β . These are different operations and have different certificates.

The results illustrate why a single statement that an expansion is “Gevrey” or “optimally truncated” can conceal important information. The index convention, lattice scale, phase, representation, and domain of uniformity all need to be specified. In particular, the improved action I_h does not contradict the location of the first formal Borel pole: the parameter-dependent coefficients and their discrete sampling remain part of the truncation problem.

11.2 What remains outside the theorem package

No theorem here supplies a universal resurgent completion of a Hahn field, a general composition theorem for exponential transseries, or a global complex Stokes classification. We do not prove that these theta coefficients belong to the repository’s original two-generator polynomial–logarithmic algebra. They naturally record a moving lattice phase and are an analytic extension of the error analysis.

The fixed-mesh result assumes $0 < h < \beta$ and excludes exact resonances. The critical-mesh result keeps λ bounded. The overlap result requires all three scale conditions in (7.3) and phase separation from resonance. The pole count and the ray are fixed. These restrictions are explicit mathematical hypotheses, not cosmetic qualifications.

11.3 A realistic formalization route

A useful first formalization target is smaller than the complete article. The finite geometric identity, positivity of W , and the partial-order comparison of truncated positive sums give a

direct certificate layer. The variance identity and increasing neighboring ratios supply a global optimization certificate without formalizing theta functions.

The next layer is real analysis: the integral and lattice tail bounds, the slope theorem, and residual-to-inverse transport. These fit the repository’s stated separation between coefficient identities and analytic remainder transport [3]. Only afterward is it necessary to formalize Gaussian lattice domination, the differentiable uniform saddle expansion, and the zero-mean tilt. The pole hierarchy is then a parameterized reuse of the same theorem rather than a separate proof for every L .

No Lean source is included because none was compiled for this work. A symbolic equality in the verification script is not described as a kernel-checked theorem.

12 Further research questions and topics

The following are precisely delimited next questions. They are not being advertised as established open problems throughout the published literature; a dedicated priority review would be needed for that claim.

- Q1. The optimized resonance boundary layer.** Determine a uniform minimum when $\lambda \rightarrow \infty$ and $\theta \rightarrow 0$ simultaneously. The entropy formula of Proposition 7.1 is not uniform there. At exact resonance an order-independent lattice contribution dominates, and subordinate poles and neighboring lattice points choose the optimal order. Find the scales at which those mechanisms exchange control.
- Q2. The complete fixed- h phase diagram for $h \geq \beta$.** There is no lattice point below the first unresolved pole in this regime. Determine the limiting optimal integer order and all transition values of h . The exact positive moment representation reduces the question to a well-defined family of kernel comparisons, but the two-point chord argument of Theorem 6.1 no longer applies unchanged.
- Q3. Coalescing ray parameters.** Allow $(a_j - a_*)x$ to stay bounded while $x \rightarrow \infty$. The multiplicity m_* should be replaced by a finite weighted collection of nearly minimal rays. Determine the accompanying changes to the tilt, bounded order correction, and exponential separation estimates, with explicit uniformity in the ray parameters.
- Q4. A growing number of resolved poles.** Let $L = L(x) \rightarrow \infty$. Identify the largest growth range in which a uniform version of the pole hierarchy remains useful. The denominator scale, saddle width, pole-separation estimate, and cost of evaluating the resolved correction must be controlled together; the present fixed- L statement does not settle this problem.
- Q5. Optimal computational cost rather than optimal truncation.** Choose K , L , the lattice cutoff, pole cutoff, and working precision to minimize total cost at a required certified accuracy. This requires a condition-number analysis of the combined kernels, not only a small mathematical remainder. A positive-kernel formulation may avoid much of the cancellation in T_K plus the resolved correction.
- Q6. A fully validated implementation.** Implement outward-rounded versions of Theorem 9.1 and Lemma 5.1, including theta-root enclosures and a robust near-tie test for adjacent truncation orders. The existing code provides reproducible diagnostics, but arithmetic certification remains a distinct engineering and proof task.

- Q7. Complex sectors and Stokes data.** Continue the positive real-axis formulas to specified complex sectors and determine which theta factors survive contour deformation. Positivity and real convexity are essential in the present optimization proof; their complex replacements need to be stated rather than assumed. Any relation to the known quantum-dilogarithm Stokes structure should be made at the level of an explicit normalization.
- Q8. Signed or oscillatory analogues.** For factorial quotients whose analogue of W changes sign, determine conditions that preserve a useful optimization principle. Leading-ray cancellation can move the action and destroy log-convexity. A criterion in terms of the signed spectral measure would extend the method beyond positive multinomial rays.
- Q9. A sharp inverse transseries, not only an enclosure.** Combine the theta-resolved residual with an all-order expansion of F'_h and its higher derivatives to obtain phase-dependent inverse-error coefficients. The present slope bound gives a reliable enclosure, but it does not yet identify the sharp leading constant for every inverted parameter regime.
- Q10. A reusable formal saddle-to-lattice theorem.** Formalize a theorem for positive sums with a moving nondegenerate saddle, a bounded scaled mesh, and a differentiable order parameter. Show how the zero-mean tilt and neighboring-ratio certificate can be exported as a common interface for other transseries in ProveIt, without constructing an unnecessarily broad formal transseries field first.

13 Conclusion

The lattice-to-integral question has two answers, depending on what is being optimized. A prescribed near-least-term rule gives an ordinary theta factor. The exact minimum over all truncation orders gives a tilted theta factor, with an explicit $\sqrt{x}$ displacement and a calculable bounded correction. At fixed nonresonant mesh, a logarithmic-mean chord produces a strictly larger action and a residual integer-order oscillation. A direct overlap theorem connects the two descriptions.

The same proof works after resolving any fixed number of partial-fraction poles. This yields a concrete, signed, and finitely certifiable hierarchy of more accurate approximations and inverse enclosures. The resulting extension is narrower than a general theorem about resurgence, but it is substantive: it determines errors and optimal orders that fixed-order formal coefficients alone do not reveal.

A An explicit coefficient-generation recurrence

For implementability, the generating rule (4.2) can be written using only polynomial arithmetic. Let

$$A(\varepsilon, u) = \frac{1}{1 + \varepsilon u + \varepsilon^2 u^2 / 2} = \sum_{n \geq 0} a_n(u) \varepsilon^n, \quad (\text{A.1})$$

$$a_0 = 1, \quad a_1 = -u, \quad a_n = -u a_{n-1} - \frac{1}{2} u^2 a_{n-2} \quad (n \geq 2), \quad (\text{A.2})$$

$$E_j(u, \eta) = \frac{(-1)^{j+3} u^{j+2}}{j+2} + \frac{\eta (-1)^{j+2} u^{j+1}}{j+1} \quad (j \geq 1), \quad (\text{A.3})$$

$$H(\varepsilon, u, \eta) = \exp \left(\sum_{j \geq 1} E_j \varepsilon^j \right) = \sum_{n \geq 0} h_n(u, \eta) \varepsilon^n. \quad (\text{A.4})$$

Differentiating H formally gives the recurrence

$$h_0 = 1, \quad nh_n = \sum_{j=1}^n jE_j h_{n-j}, \quad P_n = \sum_{j=0}^n a_j h_{n-j}. \quad (\text{A.5})$$

This proves finiteness of every coefficient and gives an elementary algorithm for arbitrarily many saddle corrections. Averaging each monomial u^j is equivalent to applying ∂_η^j to $\mathcal{J}$. Thus the all-order expansion can be represented using a finite collection of theta derivatives at every requested order.

For a bounded unscaled offset σ instead of a bounded tilt, multiply $Q_\varepsilon(u, 0)$ by $(1 + \varepsilon u)^\sigma$. This produces the endpoint-offset polynomials used in Corollary 5.3. These are coefficient identities; the analytic remainder in the resulting expansion still comes from Theorem 4.1, not from the recurrence alone.

B Claim ledger and provenance

Result	Content	Status in this article
Exact positive remainder	Bernoulli kernel, signed multinomial weight, continuous endpoint	Starting mechanism rederived; not claimed new
Theorem 4.1	All-order critical-mesh expansion with phase and order tilt, including derivatives	Developed and proved here
Theorem 5.2	Global minimization, $\sqrt{z}$ order shift, bounded order correction, first minimum correction	Developed and proved here
Corollary 5.3	Explicit $-13/24$ coefficient and integer order oscillation	Derived refinement
Theorem 6.1	Logarithmic-mean action, strict action improvement, periodic order prefactor	Developed and proved here
Theorem 7.2	Direct mesoscopic matching theorem	Developed and proved here
Theorem 8.1	Reoptimized, exactly pole-resolved hierarchy	Developed and proved here from the same kernel
Finite and inverse certificates	Positive-sum truncation and slope-based transport	Explicit reusable estimates; conditioning rederived
General resurgence, complex Stokes data, growing L	Wider analytic extensions	Not established here

The inspected earlier manuscript is identified by filename `uniform_q_multinomial_transeries.tex`, dated 29 September 2026, 1,521 source lines in its saved version. Its equation labels `eq:Rexact`, `eq:Rzero`, `eq:discrete-remainder`, and `thm:resonance`, together with Question 1 and Question 2 of its further-research section, specify the comparison. Its content was read, not inferred from its abstract alone. The separate saved manuscript on a uniform resurgent crossover and the saved one-exponential reversion manuscript were checked at the scope/abstract level to avoid presenting their principal results as new results of this article.

The repository READMEs distinguish exact formalization from partial interfaces and absent assembled analytic developments. This article preserves that distinction. It does not infer machine verification from a README inventory, and it does not assert commit-to-PDF publication parity for a repository render that was not inspected.

C Reproduction and file inventory

The package contains the editable source `article.tex`, the compiled `article.pdf`, two figure files with their plotting script, the verification script, CSV data, generated numerical tables, a JSON receipt, and a source/claim audit. From the package directory:

```
python -m pip install -r requirements.txt
python code/verify.py
python code/plots.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The shipped figures and tables allow rebuilding the PDF without rerunning the numerical scripts. The tests contain no network calls and do not modify the ProveIt repository. The manuscript has been compiled and its rendered pages inspected; that document-production check is separate from mathematical verification.

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